IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

POWER ELECTRONICS & POWER SYSTEMS

Monday 15 May 14:30

Time allowed: 2 hours

There are FIVE questions on this paper.

Answer all questions.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : M. O'Malley
Second Marker(s) : B. Pal

© Imperial College London

Information for Candidates

Switch-Mode Power Supplies

Voltage ratio equation for buck SMPS in continuous mode $\frac{V_O}{V_I} = \delta$

Voltage ratio equations for boost SMPS in continuous mode $\frac{V_O}{V_I} = \frac{1}{1-\delta}$

Voltage ratio equations for buck-boost SMPS in continuous mode $\frac{V_O}{V_I} = \frac{-\delta}{1-\delta}$

Three-Phase Systems (Balanced)

Line/Phase Voltages and Currents

Star $V_L = \sqrt{3}\, V_P;\ I_L = I_P$ Delta $V_L = V_P;\ I_L = \sqrt{3}\, I_P$

Power $P_{3\phi} = 3\, V_P I_P cos(\phi) = \sqrt{3}\ V_L I_L cos(\phi)$

Transformers

Ideal transformer $\bar{V}_2 = k_N \bar{V}_1$; $\bar{I}_2{}' = k_N \bar{I}_2$

Impedances referral $\bar{Z}'_2 = {\bar{Z}_2}/{k_N{}^2}$

Voltage regulation $\Delta V_2 = \frac{\left|V_2^{o/c}\right| - |V_2^{load}|}{\left|V_2^{o/c}\right|}$

Induction Machines

$\omega_S = {\omega_E}/{P}$; $\quad s = \frac{\omega_S - \omega_R}{\omega_S}$; $\quad T_{EM} = \frac{3\, I'^2_R\, R'_R}{\omega_R}\left(\frac{1-s}{s}\right)$

Power Flow in Lines and Cables

Cable Parameters (expressed per metre)

$R'_{LF} = \frac{1}{\sigma_C \pi\, r_C^2} + \frac{1}{\sigma_C 2\pi\, r_O\, t_O}$ $\quad L' = \frac{\mu}{2\pi} ln\left({r_O}/{r_C}\right)$ $\quad C' = \frac{2\pi\varepsilon_0\varepsilon_{RI}}{ln\left({r_O}/{r_C}\right)}$ $\quad G' = \frac{2\pi\sigma_I}{ln\left({r_O}/{r_C}\right)}$

OHL Parameters in approximate form (expressed per metre)

$R'_{LF} = \frac{2}{\sigma_C \pi\, r_C^2}$ $\quad L' = \frac{\mu}{\pi} ln\left({d}/{r_C}\right)$ $\quad C' = \frac{\pi\varepsilon_0\varepsilon_{RI}}{ln\left({d}/{r_C}\right)}$ $\quad G' = \frac{\pi\sigma_I}{ln\left({d}/{r_C}\right)}$

Power Flow (full form)

$$P_S = \frac{{V_S}^2}{Z_{SR}}\cos(\theta) - \frac{V_R V_S}{Z_{SR}}\cos(\theta+\delta) \qquad Q_S = \frac{{V_S}^2}{Z_{SR}}\sin(\theta) - \frac{V_R V_S}{Z_{SR}}\sin(\theta+\delta)$$

Power Flow assuming inductance-dominated impedance

$$P_S \simeq \frac{V_R V_S}{X_{SR}}\sin(\delta) \qquad Q_S \simeq \frac{V_S}{X_{SR}}\left(V_S - V_R\cos(\delta)\right)$$

Voltage Drop (approximate form) $\Delta V = |V_S| - |V_R| \approx \frac{R\, P_S + X\, Q_S}{|V_S|}$

1. This question addresses power flow in lines and cables.

   Consider the characteristics of single-phase overhead lines.

   a) An overhead line is found to have an inductance per unit length of 200 μH/km and is operated at 275 kV (on both the sending and receiving ends) and 50 Hz. For a line with a length of 50km, calculate the power transferred with a load angle (that is, the angle difference between the sending-end and receiving-end voltages) of 10°. [4]

   b) For the line in (a), if the current is to be limited to 2kA, calculate the maximum load angle. [3]

   c) An overhead line has a series impedance of 0.55 + j 0.40 Ω. The voltage at the sending end is 11 kV. A real power of +9.0 MW is measured flowing into the line at the sending end. The voltage at the receiving end is measured at 11kV. Estimate the reactive power as seen at the sending end. [3]

2. This question addresses induction machines.

   An induction machine is rated as follows: 415 V line; 50 Hz; 3-phase; 8-pole; star-connected. The equivalent circuit parameters, referred to the stator, are:

Stator resistance	1.0 Ω
Referred rotor resistance	0.5 Ω
Stator leakage reactance	2.0 Ω
Referred rotor leakage reactance	1.0 Ω
Magnetising reactance	350 Ω
Magnetizing resistance	350 Ω

   a) Briefly and using some quantification justify ignoring the magnetising impedance for a slip of 2 % and indicate at what speed this approximation may not be valid. [2]

   b) Ignoring the magnetizing impedance calculate the following for a slip of 2 %:

      i) The speed in revolutions per minute. [2]

      ii) The stator current in Amps. [2]

      iii) The electromagnetic torque in Nm. [2]

      iv) The power factor (stating whether it is lagging or leading). [2]

3. This question addresses transformers

   A 50 Hz step-up transformer has the following parameters for its equivalent circuit

Primary winding resistance	R_1 = 1.0 Ω
Primary leakage reactance	X_1 = 5.0 Ω
Secondary winding resistance	R_2 = 0.2 Ω
Secondary leakage reactance	X_2 = 1.0 Ω
Magnetising resistance	R_M = 5,000 Ω
Magnetising reactance	X_M = 20,000 Ω

a) The transformer primary is connected to a voltage of V_1 = 10 kV and the number of turns on one side of the transformer is 25 % the number of turns on the other side.

i) Calculate the nominal secondary voltage. [2]

ii) Calculate the voltage on the secondary side when it is open circuit. [6]

iii) Calculate the magnetizing losses and the transformer efficiency when the transformer is open circuit on the secondary side. [4]

iv) If the transformer was operated at 60 Hz give an estimate of any increase or decrease in magnetizing losses (there is no need to recalculate the losses) and indicate what physical phenomena is responsible for any change. [4]

b) Why were transformers fundamental to the outcome of the "current wars" i.e., AC versus DC in the late 19th century. [4]

4. This question addresses switch-mode power supplies.

a) A Buck SMPS is to be designed to convert 50 V to 20 V with a maximum output power of 200 W. It will operate in **continuous** conduction mode.

The following design choices have been made.

Switching frequency, f = 40 kHz

Inductor, L = 500 μH

Capacitor, C = 1,000 μF with an effective series resistance, R_{ESR} = 30 mΩ

i) Calculate duty cycle needed for the 50V to 20V voltage conversion ratio, neglecting the voltage drop on the MOSFET and diode. [2]

ii) Using the same input voltage and duty cycle as in i), calculate the output voltage if we assume a 0.7V voltage drop on the diode (still neglecting the voltage drop on the MOSFET). [4]

iii) The MOSFET to be used has the following characteristics:

Turning on Energy E_{on} 0.09mJ

Turning off Energy E_{off} 0.06mJ

On-state resistance R_{on} 50mΩ

Off-state resistance R_{off} infinite

It is required that the junction temperature is 120°C or less in operation. The temperature of the ambient air is 40°C. Calculate the maximum thermal resistance R_{th} between the ambient and the junction at the maximum output power of 200 W. [10]

iv) Calculate the power losses on the effective series resistance of the capacitance and compare that with the power losses on the MOSFET (This question may have higher difficulty than others). [4]

b) A second SMPS of Buck-Boost form is required to produce a -25 V output from the 25 V input. The circuit will be operated at a switching frequency of 40 kHz in **discontinuous** conduction mode. For operation at 200 W output power a duty cycle of 0.2 will be used.

i) Calculate the stored energy that should be present in the inductor at the end of the on-time to transfer 200 W under the stated conditions. [2]

ii) Calculate the value of inductor required. [4]

iii) Calculate the duration in a switching cycle when neither the MOSFET nor the diode is conducting. [2]

iv) Calculate the output voltage ripple across the capacitor if the effective series resistance of the capacitor is R_{ESR} = 20 mΩ (you may ignore the ripple across the capacitance of the capacitor). [2]

5. This question addresses three-phase power systems

a) What are the advantages of three phase power systems over single phase power systems. [4]

b) Consider a three-phase 60Hz balanced supply with a line voltage of 400 V. Three equal impedances are formed into a star connection and connected to the supply. The line current drawn is $200\sqrt{2/3} \angle +45^\circ \ A$. (The angle of the current is with respect to the phase voltage).

i) Calculate the real and reactive powers drawn by the impedances. [3]

ii) Calculate the impedances. [3]

iii) Calculate the value of the circuit element (inductor or capacitor) that can be placed in series with the A phase impedance so as it is purely resistive, i.e., zero reactance. [4]

iv) With the A phase impedance purely resistive (as in (iii) above) and the B & C phase impedances unchanged the load is no longer balanced. Assuming an ABC sequence and a 4-wire configuration calculate all three phase and line currents and current in the neutral. [4]

v) With the A phase impedance purely resistive (as in (iii) above) and the B & C phase impedances unchanged the load is no longer balanced. Assuming an ACB sequence and a 3-wire configuration calculate all three phase currents and the voltage at the star point. [8]

vi) For (v) above sketch the phasor diagram showing all 3 phase currents and the voltage at the star point. [4]